MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. Let A be a point on the x-axis. Common tangents are drawn from A to the curves $x^2 + y^2 = 8$ and $y^2 = 16x$. If one of these tangents touches the two curves at Q and R, then $(QR)^2$ is equal to

(1) 76 (2) 72

(3) 64 (4) 81

Answer (2)

Sol. Let a tangent on $y^2 = 16x$ be $y = mx + \frac{4}{m}$

For common to $x^2 + y^2 = 8$

$$\frac{4}{m} = 2\sqrt{2}\left(1+m^2\right)$$

$$\Rightarrow \quad \frac{2}{m^2} = 1+m^2 \quad \Rightarrow \quad m = \pm 1$$

Taking one of the tangent $y = x + 4$

Point of tangency with $y^2 = 4x$

$x^2 + 8 + 16 = 4x \quad \Rightarrow x = 4$ & $y = 8$

$\therefore \quad Q(4, 8)$

and for $x^2 + y^2 = 8$

$2x^2 + 8x + 8 = 0$

$x^2 + 4x + 4 = 8 \Rightarrow x = -2, y = 2 \Rightarrow R = (-2, 2)$

$(QR)^2 = 6^2 + 6^2$

$= 72$

62. Let f, g and h be the real valued functions defined of $\mathbb{R}$ as

$$f(x)\begin{cases}\frac{x}{|x|}, & x \neq 0\\ 1, & x = 0\end{cases}, \quad g(x)\begin{cases}\frac{\sin(x+1)}{(x+1)}, & x \neq -1\\ 1, & x = -1\end{cases}$$

and $h(x) = 2[x] - f(x)$, where $[x]$ is the greatest integer $\leq x$.

Then the value of $\lim\limits_{x\to 1} g\left(h\left(x-1\right)\right)$ is

(1) 1 (2) –1

(3) sin(1) (4) 0

Answer (1)

Sol. $f(x) = \text{sgn}(x)$

$h(x) = 2[x] - \text{sgn}(x)$

If $x \to 1^+$ then $h(x-1) = 2[x] - 2 - \text{sgn}\,(x-1)$

$= 0 - 1 = -1$

& if $x \to 1^-$ then $h(x-1) = 2[x] - 2 - \text{sgn}\,(x-1)$

$= -2 + 1 = -1$

$$\therefore \quad \lim_{x\to 1^+} g\left(h\left(x-1\right)\right) = \lim_{x\to 1^+} \frac{\sin\left(h\left(x+1\right)+1\right)}{h\left(x-1\right)+1} = 1$$

$$\lim_{x\to 1^-} g\left(h\left(x-1\right)\right) = \lim_{x\to 1^-} \frac{\sin\left(h\left(x-1\right)\right)+1)}{h\left(x-1\right)+1} = 1$$

63. Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \ldots\ldots, a_{100}$ is 25. Then S is

(1) {9} (2) {99}

(3) ϕ (4) $\mathbb{N}$

Answer (4)

Sol. Let $a_1 = a \Rightarrow a_2 = a + 1, \ldots\ldots a_{100} = a + 90$

$$\mu = \frac{\frac{100a + (99\times 100)}{2}}{100} = a + \frac{99}{2}$$

$$\text{M.D} = \frac{\sum|a_1 - \mu|}{100} = \frac{\left(\frac{99}{2} + \frac{97}{2} + \ldots\ldots \frac{1}{2}\right)^2}{100} = \frac{2500}{100} = 25$$

$\therefore \quad a \to z$

64. If a plane passes through the points $(-1, k, 0)$, $(2, k, -1)$, $(1, 1, 2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is

(1) $\frac{13}{6}$ (2) $\frac{17}{5}$

(3) $\frac{5}{17}$ (4) $\frac{6}{13}$

Answer (1)

Sol. Let $P \equiv (-1, k, 0)$, $Q \equiv (2, k, -1)$ & $R(1, 1, 2)$

$\vec{PR} = 2\hat{i} + (1-k)\hat{j} + 2\hat{k}$

& $\vec{QR} = -\hat{i} + (1-k)\hat{j} + 3\hat{k}$

$\therefore$ Normal to plane will be

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & (1-k) & 2 \\ -1 & (1-k) & 3 \end{vmatrix} = \hat{i}(1-k) - \hat{j}(8) + 3\hat{k}(1-k)$$

If line is parallel to this we have

$1(1 - k) + 1(-8) + (-3)(1 - k) = 0$

$\Rightarrow$ $2(1 - k) = -8$

$\Rightarrow$ $1 - k = -4 \Rightarrow k = 5$

$\therefore$ $\dfrac{k^2+1}{(k-1)(k-2)} = \dfrac{26}{4.3} = \dfrac{13}{6}$

65. $\lim\limits_{n\to\infty} \dfrac{3}{n}\left\{4 + \left(2+\dfrac{1}{n}\right)^2 + \left(2+\dfrac{2}{n}\right)^2 + \ldots + \left(3-\dfrac{1}{n}\right)^2\right\}$ is equal to

(1) $\dfrac{19}{3}$

(2) 12

(3) 19

(4) 0

Answer (3)

Sol. $\lim\limits_{n\to\infty} \sum\limits_{r=0}^{n-1} \dfrac{3}{n}\left(2+\dfrac{r}{n}\right)^2$

$= \int_0^1 3(2+x)^2\,dx$

$= 3 \cdot \dfrac{(2+x)^3}{3}\Big|_0^1$

$= 3^3 - 2^3 = 19$

66. Let $\vec{a}$ and $\vec{b}$ be two vectors, Let $|\vec{a}| = 1$, $|\vec{b}| = 4$ and $\vec{a}.\vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$, then the value of $\vec{b}.\vec{c}$ is

(1) –84

(2) –48

(3) –24

(4) –60

Answer (2)

Sol. $\vec{b} \cdot \vec{c} = \vec{b} \cdot (2\vec{a} \times \vec{b}) - 3\vec{b} \cdot \vec{b}$

$= 0 - 3|\vec{b}|^2$

$= -48$

67. Let $\lambda \in R$, $\vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$

If $\left((\vec{a}+\vec{b}) \times (\vec{a} \times \vec{b})\right) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$

then $\left|\lambda(\vec{a}+\vec{b}) \times (\vec{a}-\vec{b})\right|^2$ is equal to

(1) 132

(2) 136

(3) 140

(4) 144

Answer (3)

Sol. $\left(\vec{a} \times (\vec{a} \times \vec{b}) + \vec{b} \times (\vec{a} \times \vec{b})\right) \times (\vec{a} - \vec{b})$

$= \left(\vec{a}(\vec{a}\cdot\vec{b}) - \vec{b}(\vec{a}\cdot\vec{a}) + \vec{a}(\vec{b}\cdot\vec{b}) - \vec{b}(\vec{a}\cdot\vec{b})\right) \times (\vec{a} - \vec{b})$

$= (\vec{a}\cdot\vec{b})(\vec{a}\times\vec{a} - \vec{a}\times\vec{b}) - (\vec{a}\cdot\vec{a})(\vec{b}\times\vec{a} - \vec{b}\times\vec{b})$

$+ (\vec{b}\cdot\vec{b})(\vec{a}\times\vec{a} - \vec{a}\times\vec{b}) - (\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a} - \vec{b}\times\vec{b})$

$= (\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a}) - (\vec{a}\cdot\vec{a})(\vec{b}\times\vec{a})$

$+ (\vec{b}\cdot\vec{b})(\vec{b}\times\vec{a}) - (\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a})$

$= (\vec{b}\times\vec{a})(\vec{b}\cdot\vec{b} - \vec{a}\cdot\vec{a})$

$= (5\vec{b}\times\vec{a})(5 + \lambda^2 - 13 - \lambda^2)$

$= 8(\vec{a}\times\vec{b})$

$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \lambda & 2 & -3 \\ 1 & -\lambda & 2 \end{vmatrix}$

$= \hat{i}(4 - 3\lambda) - \hat{j}(2\lambda + 3) + \hat{k}(-\lambda^2 - 2)$

$\Rightarrow$ $\boxed{\lambda = 1}$

$\left|\vec{a} \times (\vec{a} - \vec{b}) + \vec{b} \times (\vec{a} - \vec{b})\right|^2$

$= \left|2(\vec{a}\times\vec{b})\right|^2 = 4.35 = 140$

68. If the functions $f(x)=\frac{x^3}{3}+2bx+\frac{ax^2}{2}$ and $g(x)=\frac{x^3}{3}+ax+bx^2, a\neq 2b$ have a common extreme point, then $a+2b+7$ is equal to

(1) 3 (2) 4

(3) 6 (4) $\frac{3}{2}$

Answer (3)

Sol. $f'(x)=x^2+2b+ax$

$g'(x)=x^2+a+2bx$

$\Rightarrow$ $x=1$ is common root

$a+2b+1=0$

69. The solution of the differential equation $\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right), y(1)=0$

(1) $\log_e|x+y|-\frac{xy}{(x+y)^2}=0$

(2) $\log_e|x+y|+\frac{2xy}{(x+y)^2}=0$

(3) $\log_e|x+y|+\frac{xy}{(x+y)^2}=0$

(4) $\log_e|x+y|-\frac{2xy}{(x+y)^2}=0$

Answer (2)

Sol. $y=\nu x$

$$\nu+x\frac{d\nu}{dx}=-\left(\frac{1+3\nu^2}{3+\nu^2}\right)$$

$$x\frac{d\nu}{dx}=-\left(\frac{1+3\nu^2}{3+\nu^2}+\nu\right)$$

$$\frac{d\nu}{dx}=-\left(\frac{(1+\nu)^3}{3+\nu^2}\right)$$

$$\Rightarrow \frac{3+\nu^2}{(1+\nu)^3}=\frac{-dx}{x}$$

$$\Rightarrow \ln|\nu+1|+\frac{2\nu}{(\nu+1)^2}=C-\ln|x|$$

$x=1,\ \nu=0 \Rightarrow C=0$

$$\Rightarrow \ln|x+y|-\ln|x|+\frac{2xy}{(x+y)^2}=-\ln|x|$$

70. The parabolas : $ax^2+2bx+cy=0$ and $dx^2+2ex+fy=0$ intersect on the line $y=1$.

If a, b, c, d, e, f are positive real numbers and a, b, c are in G.P., then

(1) $\frac{d}{a},\frac{e}{b},\frac{f}{c}$ are in A.P. (2) d, e, f are in G.P.

(3) d, e, f are in A.P. (4) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

Answer (1)

Sol. Let point of intersection be $(\alpha, 1)$

$\alpha x^2+2b\alpha+c=0$ ...(i)

and $d\alpha^2+2e\alpha+f=0$ ...(ii)

$\Rightarrow$ $a\alpha^2+2\sqrt{ac}\alpha+c=0$ $(\because b^2=ac)$

$$\left(\sqrt{a}\alpha+\sqrt{c}\right)^2=0$$

$$\alpha=-\sqrt{\frac{c}{a}}$$

Put the value α in (ii),

$$d\frac{c}{a}-2e\sqrt{\frac{c}{a}}+f=0$$

$$\frac{d}{a}-\frac{2e}{\sqrt{ac}}+\frac{f}{c}=0$$

$$\frac{d}{a}+\frac{f}{c}=2\frac{e}{b}$$

$\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

71. Consider the following statements:

P : I have fever

Q : I will not take medicine

R : I will take rest

The statement "If I have fever, then I will take medicine and I will take rest"

(1) $(P\vee\sim Q)\wedge(P\vee\sim R)$

(2) $((\sim P)\vee\sim Q)\wedge((\sim P)\vee\sim R)$

(3) $((\sim P)\vee\sim Q)\wedge((\sim P)\vee R)$

(4) $(P\vee Q)\wedge((\sim P)\vee R)$

Answer (3)

Sol. The given expression is

$P\rightarrow\sim Q\wedge R$

$\equiv(\sim P)\vee(\sim Q\wedge R)$

$\equiv(\sim P\vee\sim Q)\wedge(\sim P\vee R)$

72. Let $a_1 = 1, a_2, a_3, a_4, \ldots$ be consecutive natural numbers. Then

$$\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+\ldots.$$

$$+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right) \text{ is equal to}$$

(1) $\tan^{-1}(2022)-\frac{\pi}{4}$ (2) $\cot^{-1}(2022)-\frac{\pi}{4}$

(3) $\frac{\pi}{4}-\cot^{-1}(2022)$ (4) $\frac{\pi}{4}-\tan^{-1}(2022)$

Answer (1)(3)*

Sol. $\tan^{-1}(a_2)-\tan^{-1}(a_1)+\tan^{-1}(a_3)-\tan^{-1}(a_2)$

$$+\ldots+\tan^{-1}a_{2022}-\tan^{-1}a_{2021}$$

$$=\tan^{-1}(a_{2022})-\tan^{-1}(a_1)$$

$$=\tan^{-1}(2022)-\frac{\pi}{4}$$

73. Let $x=\left(8\sqrt{3}+13\right)^{13}$ and $y=\left(7\sqrt{2}+9\right)^{9}$. If $[t]$ denotes the greatest integer $\le t$, then

(1) $[x]$ is odd but $[y]$ is even

(2) $[x] + [y]$ is even

(3) $[x]$ and $[y]$ are both odd

(4) $[x]$ is even but $[y]$ is odd

Answer (2)

Sol. If $I_1+f=(8\sqrt{3}+13)^{13}$, $f'=\left(8\sqrt{3}-13\right)^{13}$

$I_1+f-f'=$ Even

$I_1=$ Even

$I_2+f-f'=\left(7\sqrt{2}+9\right)^{9}+\left(7\sqrt{2}-9\right)^{9}$

$=$ Even

$I_2=$ Even

74. For $\alpha,\beta\in\mathbb{R}$, suppose the system of linear equations

$x - y + z = 5$

$2x + 2y + \alpha z = 8$

$3x - y + 4z = \beta$

Has infinitely many solutions. Then α and β are the roots of

(1) $x^2 + 14x + 24 = 0$ (2) $x^2 - 18x + 56 = 0$

(3) $x^2 - 10x + 16 = 0$ (4) $x^2 + 18x + 56 = 0$

Answer (2)

Sol. $\Delta=\begin{vmatrix}1 & -1 & 1\\ 2 & 2 & \alpha\\ 3 & -1 & 4\end{vmatrix}=0$

$\Rightarrow \alpha = 4$

$\Delta_3 = 0$

$$=\begin{vmatrix}1 & -1 & 5\\ 2 & 2 & 8\\ 3 & -1 & \beta\end{vmatrix}=0$$

$\Rightarrow \beta = 14$

$\therefore\ x^2 - 18x + 56 = 0$

75. The number of ways of selecting two numbers a and b, $a\in\{2,4,6,\ldots..,100\}$ and $b\in(1,3,5,\ldots.,99)$ such that 2 is the remainder when $a + b$ is divided by 23 is

(1) 108

(2) 186

(3) 54

(4) 268

Answer (1)

Sol. $a + b = 23\lambda + 2$

$\lambda = 0, 1, 2, \ldots$, but λ cannot be even as $a + b$ is odd

$\therefore\ \lambda = 1$ $(a, b) \to 12$ pairs

$\lambda = 3$ $(a, b) \to 35$ pairs

$\lambda = 5$ $(a, b) \to 42$ pairs

$\lambda = 7$ $(a, b) \to 19$ pairs

$\lambda = 9$ $(a, b) \to 0$ pairs

$\vdots$

Total = 12 + 35 + 42 + 19 = 108

76. If P is a 3×3 real matrix such that $P^T = aP + (a-1)I$, where $a > 1$, then

(1) $|\text{Adj}\,P|=\frac{1}{2}$

(2) $|\text{Adj}\,P|>1$

(3) P is a singular matrix

(4) $|\text{Adj}\,P|=1$

Answer (4)

Sol. Let $P=\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

Given : $P^T = aP + (a-1)I$

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = \begin{bmatrix} aa_1+a-1 & ab_1 & ac_1 \\ aa_2 & ab_2+a-1 & ac_2 \\ aa_3 & ab_3 & ac_3+a-1 \end{bmatrix}$$

$\Rightarrow \quad a_1 = aa_1 + a - 1 \Rightarrow a_1(1-a) = a-1 \Rightarrow a_1 = -1$

Similarly, $a_1 = b_2 = c_3 = -1$

Now, $\left.\begin{matrix} a_2 = ab_1 \\ b_1 = aa_2 \\ c_1 = aa_3 \end{matrix}\right] \rightarrow a_2 = a^2 a_2 \Rightarrow a_2 = 0 \Rightarrow b_1 = 0$

Similarly all other elements will also be 0

$a_2 = a_3 = b_1 = b_3 = c_1 = c_2 = 0$

$\therefore \quad P=\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$|P| = -1$

$|\text{Adj}(P)|_{n\times n} = |A|^{(n-1)}$

$\Rightarrow \quad |\text{Adj}(P)| = (-1)^2 = 1$

77. A vector $\vec{v}$ in the first octant is inclined to the x- axis at 60°, to the y-axis at 45° and to the z-axis at an acute angle. If a plane passing through the points $\left(\sqrt{2},-1,1\right)$ and (a, b, c) is normal to $\vec{v}$, then

(1) $\sqrt{2a}+b+c=1$ (2) $a+\sqrt{2b}+c=1$

(3) $\sqrt{2a}-b+c=1$ (4) $a+b+\sqrt{2c}=1$

Answer (2)

Sol. $l=\frac{1}{2}, m=\frac{1}{\sqrt{2}}, n=\cos\theta$

$l^2+m^2+n^2=1$

$\Rightarrow \quad \frac{1}{4}+\frac{1}{2}+n^2=1 \Rightarrow n^2=\frac{1}{4} \Rightarrow n=\pm\frac{1}{2}$

θ is acute $\therefore \quad n=\frac{1}{2}$

$\therefore \quad \vec{v}=k\left(\frac{1}{2}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{2}\hat{k}\right), k\in R$

$\vec{v}\cdot(\vec{a}-\vec{b})=0$

$\left(\sqrt{2}-a\right)\frac{1}{2}+(-1-b)\frac{1}{\sqrt{2}}+(1-c)\frac{1}{2}=0$

$\Rightarrow \quad \frac{a}{2}+\frac{b}{\sqrt{2}}+\frac{c}{2}=\frac{1}{2}$

$\Rightarrow \quad a+\sqrt{2}b+c=1$

$\therefore$ Option (2) is correct.

78. Let a, b, $c > 1$ a^3, b^3 and c^3 be in A.P., and $\log_a b$, $\log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an A.P., whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is –444, then abc is equal to:

(1) $\frac{343}{8}$

(2) $\frac{125}{8}$

(3) 343

(4) 216

Answer (4)

Sol. $2b^3 = a^3 + c^3$

$\left(\frac{\log a}{\log c}\right)^2 = \left(\frac{\log b}{\log a}\right)\left(\frac{\log c}{\log b}\right)$

$\Rightarrow \quad (\log a)^3 = (\log c)^3$

$\Rightarrow \quad \log a = \log c$

$\Rightarrow \quad a = c$

$\Rightarrow \quad a = b = c$

$T_1 = 2a, \; d = -\frac{3a}{5}$

$S_{20} = -444$

$\Rightarrow \quad \frac{20}{2}\left(2(2a)+(19)\left(-\frac{3a}{5}\right)\right) = -444$

$\Rightarrow \quad 10\frac{(20a-57a)}{5} = -444$

$\Rightarrow \quad 37a = 222$

$\Rightarrow \quad a = 6$

$\Rightarrow \quad abc = (6)^3 = 216$

79. The range of the function $f(x)=\sqrt{3-x}+\sqrt{2+x}$ is:

(1) $\left[\sqrt{5},\sqrt{10}\right]$ (2) $\left[\sqrt{2},\sqrt{7}\right]$

(3) $\left[\sqrt{5},\sqrt{13}\right]$ (4) $\left[2\sqrt{2},\sqrt{11}\right]$

Answer (1)

Sol. $f(x)=\sqrt{3-x}+\sqrt{x+2}$

$$y'=\frac{-1}{2\sqrt{3-x}}+\frac{1}{2\sqrt{x+2}}=0$$

$$\Rightarrow \quad \sqrt{x}+2=\sqrt{3}-x$$

$$\Rightarrow \quad x=\frac{1}{2}$$

$$y\left(\frac{1}{2}\right)=\sqrt{\frac{5}{2}}+\sqrt{\frac{5}{2}}=\sqrt{10}$$

y_{min} at $x=-2$ or $x=3$, i.e., $\sqrt{5}$

$\therefore \quad f(x)\in\left[\sqrt{5},\sqrt{10}\right]$

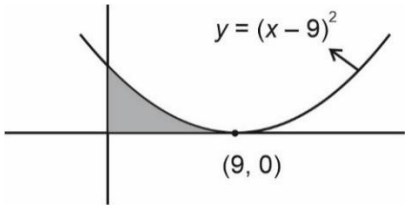
80. Let q be the maximum integral value of p in [0, 10] for which the roots of the equation $x^2-px+\frac{5}{4}p=0$ are rational. Then the area of the region $\left\{(x,y):0\le y\le(x-q)^2,0\le x\le q\right\}$ is

(1) $\frac{125}{3}$ (2) 164

(3) 243 (4) 25

Answer (3)

Sol. Given equation : $4x^2-4px+5p=0$



for rational roots, D must be perfect square

$D=16p^2-4\times4\times5p=16p(p-5)$

So, max. Integral value of $p=9$ for making D is perfect square

$\therefore \quad q=9$

Area of shared region

$$=\int_0^9(x-9)^2\,dx$$

$$=\frac{(x-9)^3}{3}\Bigg|_0^9=\frac{9^3}{3}=243 \text{ sq. units}$$

SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

81. If $\int\sqrt{\sec 2x-1}\,dx=\alpha\log_e$

$$\left|\cos 2x+\beta+\sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|+\text{ constant,}$$

then $\beta-\alpha$ is equal to _____.

Answer (01)

Sol. $I=\int\sqrt{\sec 2x-1}dx\int\sqrt{\frac{1-\cos 2x}{\cos 2x}}dx$

$$=\int\sqrt{\frac{2\sin^2 x}{2\cos^2 x-1}}dx$$

Let $\sqrt{2\cos}x=t$

$$-\sqrt{2}\sin x\,dx=dt$$

$$I=\int-\frac{dt}{\sqrt{t^2-1}}=-\ln\left|t+\sqrt{t^2-a^2}\right|+c$$

$$=-\ln\left|\sqrt{2}\cos x+\sqrt{2\cos^2}x-1\right|+c$$

$$=-\frac{1}{2}\ln\left|2\cos^2 x+\cos 2x+2\sqrt{2}\sqrt{\cos 2x.\cos^2 x}\right|+c$$

$$=-\frac{1}{2}\ln\left|2\cos 2x+1+2\sqrt{\cos 2x(1+\cos 2x)}\right|+c$$

$$=-\frac{1}{2}\ln\left|\cos 2x+\frac{1}{2}+\sqrt{\cos 2x(1+\cos 2x)}\right|+c$$

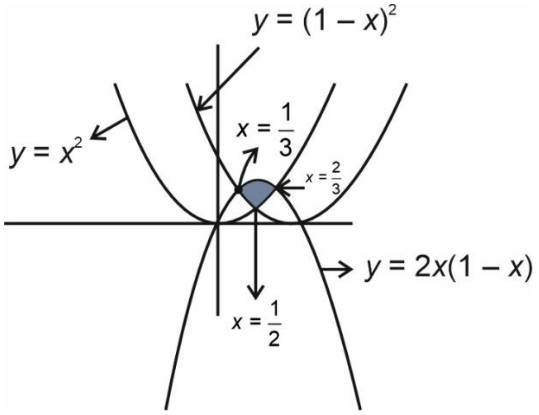
$\therefore \quad \alpha=\frac{-1}{2},\beta=\frac{1}{2}$

$\therefore \quad \beta-\alpha=1$

82. Let A be the area of the region $\left\{(x,y):y\ge x^2,y\ge(1-x)^2,y\le 2x(1-x)\right\}$. Then 540 A is equal to _____.

Answer (25)

Sol.



$y = x^2$ and $y = 2x(1 - x)$

$\Rightarrow \quad x^2 = 2x - 2x^2$

$\Rightarrow \quad x = 0, x = \frac{2}{3}$

Now,

$y = (1 - x)^2$ and $y = 2x(1 - x)$

$\Rightarrow \quad 1 + x^2 - 2x = 2x - 2x^2$

$\Rightarrow \quad 3x^2 - 4x + 1 = 0$

$x = 1, x = \frac{1}{3}$

$$\therefore \quad A = \int_{\frac{1}{3}}^{\frac{2}{3}} (2x - 2x^2)\,dx - \left\{ \int_{\frac{1}{3}}^{\frac{1}{2}} (1-x)^2\,dx + \int_{\frac{1}{2}}^{\frac{2}{3}} x^2\,dx \right\}$$

$$= \left(x^2 - \frac{2x^3}{3} \right)_{\frac{1}{3}}^{\frac{2}{3}} - \left\{ \frac{(x-1)^3}{3} \Big|_{\frac{1}{3}}^{\frac{1}{2}} + \frac{x^3}{3} \Big|_{\frac{1}{2}}^{\frac{2}{3}} \right\}$$

$$= \frac{5}{108}$$

$\therefore \quad 540A = 25$

83. 50th root of a number x is 12 and 50th root of another number y is 18. Then the remainder obtained on dividing $(x + y)$ by 25 is _____.

Answer (23)

Sol. Given $x^{\frac{1}{50}} = 12 \Rightarrow x = 12^{50}$

$y^{\frac{1}{50}} = 18 \Rightarrow y = 18^{50}$

$12 \equiv 13 \text{ (Mod 25)}$

$12^2 \equiv 19 \text{ (Mod 25)}$

$12^3 \equiv -3 \text{ (Mod 25)}$

$12^9 \equiv -2 \text{ (Mod 25)}$

$12^{10} \equiv -1 \text{ (Mod 25)}$

$12^{50} \equiv -1 \text{ (Mod 25)} \quad \ldots(i)$

Now

$18 \equiv 7 \text{ (Mod 25)}$

$18^2 \equiv -1 \text{ (Mod 25)}$

$18^{50} \equiv -1 \text{ (Mod 25)} \quad \ldots(ii)$

$\therefore \quad 12^{50} + 18^{50} \equiv -2 \text{ (Mod 25)}$

$\equiv 23 \text{ (Mod 25)}$

$\therefore \quad$ Answer = 23

84. The number of seven digits odd numbers, that can be formed using all the seven digits 1, 2, 2, 2, 3, 3, 5 is _____.

Answer (240)

Sol. $\ldots\ldots 1 \rightarrow \frac{6!}{2!3!} = 60$

$\ldots\ldots 3 \rightarrow \frac{6!}{3!} = 120$

$\ldots\ldots 5 \rightarrow \frac{6!}{3!2!} = 60$

Total = 240

85. Let $A = \{1, 2, 3, 5, 8, 9\}$. Then the number of possible functions $f : A \rightarrow A$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to _____.

Answer (432)

Sol. $f(1.n) = f(1).f(n) \Rightarrow f(1) = 1.$

$f(3.3) = (f(3))^2$

Hence the possibilities for $(f(3), f(9))$ are (1, 1) and (3, 9).

Other three *i.e.* $f(2)$, $f(5)$, $f(8)$

Can be chosen in 6^3 ways.

Hence total number of functions

$= 6^3 \times 2$

$= 432$

86. Let a line L pass through the point $P(2, 3, 1)$ and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L from the point (5, 3, 8) is α, then $3\alpha^2$ is equal to _____.

Answer (158)

Sol. $L: \frac{x-2}{1}=\frac{y-3}{-1}=\frac{z-1}{-1}=\lambda$

Any point on L can be taken as

$B(\lambda+2, -\lambda+3, -\lambda+1)$

Let $A(5, 3, 8)$

So, $AB \cdot \left(\hat{i}-\hat{j}-\hat{k}\right)=0$

$[(\lambda-3)\hat{i}-\lambda\hat{j}-(\lambda+7)\hat{k}]\cdot[\hat{i}-\hat{j}-\hat{k}]=0$

$\lambda-3+\lambda+\lambda+7=0$

$\therefore \quad \lambda=\frac{-4}{3}$

$\overrightarrow{AB}=\frac{13}{3}\hat{i}+\frac{4}{3}\hat{i}-\frac{17}{3}\hat{k}$

$|\overrightarrow{AB}|=\sqrt{\frac{169}{9}+\frac{16}{9}+\frac{289}{9}}$

$=\frac{\sqrt{474}}{3}=\alpha$

$3\alpha^2=\frac{474}{9}\times 3=158$

87. If the value of real number $\alpha > 0$ for which $x^2 - 5\alpha x + 1 = 0$ and $x^2 - \alpha x - 5 = 0$ have a common real root is $\frac{3}{\sqrt{2\beta}}$ then β is equal to _____.

Answer (13)

Sol. $x^2 - 5\alpha x + 1 = 0 \quad \ldots(1)$

$x^2 - \alpha x - 5 = 0 \quad \ldots(2)$

have a common root.

Subtracting (1) with (2) we'll get $x=\frac{6}{4\alpha}$.

Substituting in (1)

$\frac{36}{16\alpha^2}-\frac{30}{4}+1=0$

$\Rightarrow \quad \alpha^2=\frac{9}{26}$

$\alpha=\frac{3}{\sqrt{2\times 13}}$

$\therefore \quad \beta = 13$

88. The 8th common term of the series

$S_1 = 3 + 7 + 11 + 15 + 19 + \ldots,$

$S_2 = 1 + 6 + 11 + 16 + 21 + \ldots,$

is _____.

Answer (151)

Sol. First common term is 11

Common difference of series of common terms is LCM (4, 5) = 20

$a_8 = a + 7d$

$= 11 + 7 \times 20 = 151$

89. A bag contains six balls of different colours. Two balls are drawn in succession with replacement. The probability that both the balls are of the same colour is p. Next four balls are drawn in succession with replacement and the probability that exactly three balls are of the same colour is q. If $p : q = m : n$, where m and n are coprime, then $m + n$ is equal to _____.

Answer (14)

Sol. $p=\frac{6}{36}=\frac{1}{6}$

$q=\frac{{}^6C_1\times{}^5C_1\times\frac{4!}{3!}}{6^4}=\frac{120}{1296}=\frac{5}{54}$

$\frac{p}{q}=\frac{\frac{1}{6}}{\frac{5}{54}}=\frac{54}{6\times 5}=\frac{9}{5}=\frac{m}{n}$

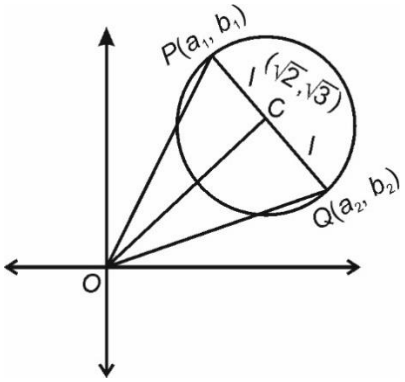
$m + n = 14$

90. Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C\left(\sqrt{2},\sqrt{3}\right)$. Let O be the origin and OC be perpendicular to both CP and CQ. If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$, then $a_1^2 + a_2^2 + b_1^2 + b_2^2$ is equal to _____.

Answer (24)

Sol. $OC \perp CP$ and $OC \perp CQ$

$\Rightarrow$ PCQ is a straight line



$$OC = \sqrt{\left(\sqrt{2}\right)^2 + \left(\sqrt{3}\right)^2} = \sqrt{5}$$

Let $CP = CQ = l$

$$[OCP] = \frac{1}{2} \times OC \times l = \frac{\sqrt{35}}{2}$$

$$l = \sqrt{7}$$

$$OP = OQ = \sqrt{(OC)^2 + l^2} = \sqrt{5+7} = \sqrt{12}$$

$$a_1^2 + a_2^2 + b_1^2 + b_2^2 = \left(a_1^2 + b_2^2\right) + \left(a_2^2 + b_2^2\right)$$

$$OP^2 + OQ^2 = 12 + 12 = 24$$

❑ ❑ ❑